# Exploration of Diffie-Hellman Permutations

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Given a prime integer $p \geq 11$, we consider the family of permutations $\tau_{p,b,c}$ parameterized by a primitive root $b$ modulo $p$ and an integer $c$ considered modulo $p-1$. The construction of this family is based on the simple observation that two different sets $\mathbb{Z}/(p-1)\mathbb{Z}$ and $(\mathbb{Z}/p\mathbb{Z})^{\times}$ can be naturally identified with the set $\{1, 2, \ldots, p-1\}$ and the latter set can be identified with $\{0, 1, \ldots, p-2\}$. See [2] and [6].

For $x \in \{0, 1, 2, \ldots, p-2\}$, we set

$$\tau_{p,b,c}(x) := \mathrm{rem}(\mathrm{rem}(b^x, p) + c, (p-1)),$$

where $\mathrm{rem}(m, N)$ denotes the remainder of division of $m$ by $N$.

Clearly, the permutation $\tau_{p,b,c}$ can be identified with the bijection $\mathbb{Z}/(p-1)\mathbb{Z} \xrightarrow{\simeq} \mathbb{Z}/(p-1)\mathbb{Z}$ defined by the formula

$$\tau_{p,b,c}(\overline{x}) := (\mathrm{rem}(b^x, p) + c) + (p-1)\mathbb{Z},$$

where $\overline{x}$ denotes the residue class of $x$ modulo $(p-1)$.

Due to the natural connection of these permutations to the Discrete Logarithm Problem (DLP), we call the permutations $\tau_{p,b,c}$ the **Diffie-Hellman permutations** or simply the DH permutations.

Setting $c = -1$, we get the family of permutations explored in paper [2] by S.V. Breslavets. Setting $c = 0$, we get the family of permutation studied in paper [6] by P. Landrock and the latter permutations are related to the Brizolis conjecture [5, Section F9]. This conjecture can be formulated as follows: given a prime $p > 3$, there exists a primitive root $b$ modulo $p$ such that the permutation $\tau_{p,b,0}$ has at least one fixed point. This conjecture was proved in paper [7] by M. Levin, C. Pomerance and K. Soundararajan.

In this brief note, we describe several open questions related to DH permutations.

**Search for unlocking sequences.** Let $b_0$ be a primitive root modulo a prime $p$. Solving the equation

$$y = \tau_{p,b_0,-1}(x)$$

for a given integer $0 \leq y \leq p-2$ is equivalent to the DLP. Here is the first natural question about the DH permutations: are there sequences $(b_1, c_1), \ldots, (b_n, c_n)$ (with each $b_j$ being a primitive root modulo $p$) such that the permutation

$$\tau_{p,b_n,c_n} \circ \cdots \circ \tau_{p,b_1,c_1} \circ \tau_{p,b_0,-1}$$

has a large number of fixed points? If yes, then can we identify such sequences?

**Unlocking sequences for the DLP in other set-ups.** Let $G$ be a cyclic group of prime order $p$ and we chose a bijection from $G$ to the set $\{0, 1, \ldots, p-1\}$. (It is natural to assume that 0 corresponds to the identity element.)

In practice, $G$ may be a subgroup of the multiplicative group of a finite field or a subgroup of the group of points of an elliptic curve (defined over a finite field). The assumption about the order of $G$ is motivated by the arguments given in [8, Section 2].

Since the set $G$ is identified with $\{0, 1, \ldots, p-1\}$, every generator $g_0$ of $G$ gives us a bijection

$$\delta_{g_0} : \{0, 1, \ldots, p-1\} \xrightarrow{\simeq} \{0, 1, \ldots, p-1\}. \tag{1}$$

Solving the equation

$$y = \delta_{g_0}(x)$$

for a given $y$ is the DLP for $G$.

Let us identify the symmetric group $S_{\{0,\ldots,p-2\}}$ with the subgroup of $S_{\{0,\ldots,p-1\}}$ in the obvious way. Then we can ask whether there are sequences $(b_1, c_1), \ldots, (b_n, c_n)$ (with each $b_j$ being a primitive root modulo $p$) such that the permutation

$$\tau_{p,b_n,c_n} \circ \cdots \circ \tau_{p,b_1,c_1} \circ \delta_{g_0}$$

has a large number of fixed points.

More generally, are there sequences $(b_1, c_1), \ldots, (b_n, c_n)$ (with each $b_j$ being a primitive root modulo $p$) for which it is easier to solve the equation

$$y = \tau_{p,b_n,c_n} \circ \cdots \circ \tau_{p,b_1,c_1} \circ \delta_{g_0}(x)$$

for a large number of integers $y$?

**A large safe prime gives us a putative pre-image resistant function.** Recall [2, Definition 2.1] that an integer $1 \leq x \leq p-2$ is called a vulnerable exponent relative to a prime $p$ if there exists a primitive root $b$ modulo $p$ such that $\tau_{p,b,-1}(x) = x$.

Let $p$ be a safe prime (i.e. $p = 2q+1$, where $q$ is also prime). According to [2, Theorem 3.5], for large safe primes $p$, roughly a quarter of integers in the set $\{1, 2, \ldots, p-2\}$ are vulnerable exponents.

Let $p$ be a safe prime $\geq 11$. Due to [2, Corollary 3.2], for every vulnerable exponent $x$, there exists a unique primitive root $b_x$ such that $\tau_{p,b_x,-1}(x) = x$. For every vulnerable exponent $x$, the corresponding primitive root $b_x$ can be found relatively fast (see the proof of [2, Theorem 2.3]). Question 4.4 in [2] is about the function that sends a vulnerable exponent $x$ to the corresponding primitive root $b_x$: is this a pre-image resistant function?

Note that the ability to factor large integers fast does not seem to help finding the pre-image of this function.

**Do DH-permutations behave like random permutations?** For every prime $p \geq 5$, the number of primitive roots modulo $p$ is $\phi(p-1)$, where $\phi$ is Euler's phi-function. Thus, if $p$ is

large, then number $\phi(p-1)$ can also be large. For example, for every safe prime $p = 2q + 1$ the number of primitive roots modulo $p$ is $q - 1$.

Let $\mathsf{PR}(p)$ be the set of primitive roots modulo $p$ and $c \in \{0, 1, \ldots, p-2\}$. We consider the set $S_{\{0,1,\ldots,p-2\}}$ with the uniform distribution. Is this true that, for large primes $p$, $\{\tau_{p,b,c}\}_{b\in\mathsf{PR}(p)}$ is close to a random sample of permutations in $S_{\{0,1,\ldots,p-2\}}$?

A supporting evidence for this statement is helpful for addressing Question 4.4 in [2] mentioned above.

Many useful statements about random permutations can be found in [1].

## References


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